# Optimized Legal Text Summarization: An FFT and PSO-based approach

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**Abstract**

**Extracting important details from long text data, especially legal documents, is difficult because these texts are complex and very long. Legal texts can have a lot of words, which makes it important to find ways to summarize them that are both accurate and make sense in context. This study presents a new method that combines three main parts: TF-IDF to assess how important sentences are, PSO to pick the most useful sentences, and FFT to improve the flow and clarity of the summary. This method is mainly designed for legal documents but was also tested on other types of long texts. TF-IDF gives scores to sentences based on how relevant their words are. PSO is used to select the best sentences from the text. FFT helps make the summary more natural by looking at patterns in how often words are used. The method was tested on three different sets of data—MildSum with 3122 documents (average time: 12 seconds), ILC with over 3000 articles (17. 62 seconds), and DUC 2004 with 500 articles (26. 84 seconds). The results show that this new method works better and faster than traditional methods. By using a mix of statistical, optimization, and signal processing techniques, this approach makes it easier to summarize large legal texts, helping legal professionals and researchers find information more quickly.**

**Keywords: extractive text summarization, TF-IDF, Particle Swarm Optimization, Fast Fourier Transform, MildSum, ILC, DUC2004, ROUGE scores.**

## I Introduction

Text summarization is a fundamental task within the field of natural language processing (NLP), focused on producing shortened versions of lengthy documents while retaining their essential meaning and key information. With the exponential rise in textual data across diverse sectors, the need for effective summarization techniques has become more critical for tasks such as knowledge extraction, decision-making, and efficient information retrieval. Legal text summarization is one of the most challenging domains in this regard. Legal documents like court decisions, contracts, case files, and laws, are usually quite long, hard to read with inclusion of a lot of technical words. Therefore, it can be challenging to generate efficient summaries from them.

Generating summaries of legal texts happens to be very important for Lawyers, researchers, and judges who often have to go through a lot of documents to find the key facts, arguments, and rulings. Since these documents are so long and complicated, reading them completely takes a lot of time and effort. Summarisation can help reduce this work. Using automatic tools for summarisation and hence, extracting the most important information can help legal professionals save time and focus on what's really important. This can also help with legal research, writing reports, and managing legal information better.

There are two main types of Summarization methods: extractive and abstractive. Extractive methods select the most important sentences from the text document without altering their meaning, using techniques like TF-IDF, LSA, and models like BERT or GPT. On the other hand, Abstractive methods rephrase the original text by generating

new sentences. Abstractive methods might sound more natural, however, they sometimes fail to make much sense. It is extremely important to keep the original meaning and context intact when we talk about legal texts, and hence, extractive methods are usually preferred.

This research introduces an integrated form of extractive methods that is designed specifically for legal texts. It optimizes the traditional statistical techniques. Fast Fourier Transform is responsible for finding the key parts of the text. Then, particle swarm optimization is used to select the best sentences based on these features. And finally, a threshold is set to select the top-K sentences to form the summary. This approach focuses on generating useful summaries while keeping the original meaning of the text intact.

We tested our method using the DUC 2004 dataset, a big collection of different legal documents, along with MildSUM and ILC dataset. These datasets are complex and varied, making them a good test for evaluating summarization methods. Further results show that our model performs well with different types and lengths of legal documents making it suitable for handling large amounts of legal text in the real-world.

The major areas that this study focuses on:

(1) A new summarization system is introduced that uses Fast Fourier Transform (FFT), TF-IDF, and Particle Swarm Optimization (PSO) together to pick the most important sentences in legal texts.

(2) This approach does not require labeled data, which means it does not require a lot of supervised training and can be used more widely in legal situations in the real-world.

(3) We evaluated our method on vast and complex legal datasets, proving that it is reliable, fast, and able to handle documents with a lot of words and complex terminology.

# II Preliminaries

## *i TF-IDF (Term Frequency-Inverse Document Frequency)*

TF-IDF is a statistical method widely used to determine how important a word is in a document compared to a larger set of documents. It comprises of two parts: term frequency (TF), which represents how often a word appears in a document, and inverse document frequency (IDF), which gives less weight to words that appear often in many documents and vice-versa. Both of these parts together help to identify the words that are unique to a particular text and signifies the importance of that word. TF-IDF is a useful method for identifying important parts of text during summarization.

## *ii Fast Fourier Transform (FFT)*

The Fast Fourier Transform (FFT) is a strong algorithm that quickly calculates the Discrete Fourier Transform (DFT) and its reverse. It helps break down a signal into different frequencies, which is very helpful in areas like signal and image processing, removing noise, and finding patterns.

Calculating the DFT directly is slow for big data, needing $O(n^2)$ operations. But FFT makes this faster, with only $O(n \log n)$ operations, by using math tricks. Algorithms like Cooley-Tukey and Split-Radix make FFT even more efficient by splitting large DFT problems into smaller ones, which speeds things up and uses resources more wisely.

## *iii Power Spectrum Analysis*

Power spectrum analysis[2] focuses on identifying how energy or signal strength is distributed across various frequency bands within a dataset. In the context of textual data, it helps examine variations in features such as term distributions or n-gram frequencies. This analysis can reveal prominent patterns and structural characteristics in the document, aiding the summarization process by highlighting areas of higher informational content.

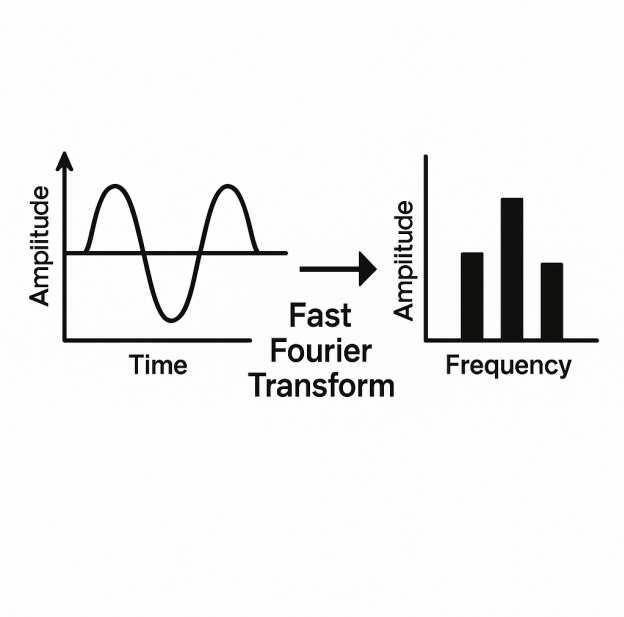



Figure 1: Visualization of Fast Fourier Transform (FFT).

### *iv Particle Swarm Optimization (PSO)*

PSO is a type of computational method that mimics natural events like how birds fly in groups or fish swim together. In this method, each part, called a particle, stands for a possible answer. These particles move through a space looking for the best answer, using what they have learned before and what the whole group has learned. Their movement depends on two main things: where they have done best on their own and where the whole group has done best. PSO works well for problems that need to find the best answer in a continuous or broken way because it's easy to use and can be changed to fit different needs.

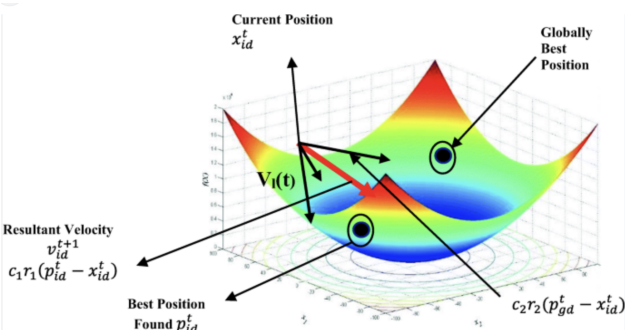



Figure 2: PSO Particles Converging Towards the Optimal Solution.

## III Related Work

Many Researchers across the world have offered different ways to handle the unique language and structure challenges in the legal documents. Kore et al. (2020)[3] integrated traditional machine learning models with natural language processing (NLP) methods generating summaries from legal documents across different types of data. Jain et al. (2021)[4] reviewed summarization methods in legal areas, determining current limitations and proposing future research ideas. Anand and Wagh (2022)[5] worked with deep learning models specifically designed for legal document summarization, leading to good results and keeping the legal meaning intact. Ghosh et al.

(2022)[6] targetted Indian legal content, developing a framework through normalization methods to better handle legal terms and sentence structures.

Takale et al.(2022)[7] proposed a rule-based approach using a Ripple Down Rules system for domain-specific summarization, which was later improved in 2023 with better knowledge representation methods designed for legal texts. Koniaris et al. (2023)[8] tested summarization models on Greek(non-english) legal cases, providing important insights into challenges in multilingual summarization. Sharma et al. (2023)[9] showed how summarization models have evolved in Indian legal documents world.

Initially, Luhn (1958)[10] and Edmundson (1969)[11] developed early statistical methods for text summarization, which focused on word frequency. Later, Mihalcea and Tarau's TextRank (2004)[12] introduced more structured methods which used graph-based sentence ranking. Nallapati et al. (2016)[13] introduced sequence-to-sequence models leading to big improvements in abstractive summarization and evaluation became more standardized thanks to Lin's ROUGE metric (2004)[14]. See et al. (2017)[15] and Paulus et al. (2018)[16] looked into ways to improve coherence and generation in summaries.

Transformer-based models are the major reason of the recent progress. BERT (Devlin et al. , 2018)[17] and GPT (Radford et al. , 2018)[18] greatly improved both extractive and abstractive summarization. Researchers like Liu et al. (2020)[19] and Zhang et al. (2021)[20] applied the summarization techniques to multimodal content. In extractive summarization, Krishnan et al. (2019)[21] created a supervised method that achieved strong ROUGE scores. Khan et al. (2019)[22] used TF-IDF and K-means for better sentence clustering. Kryscinski et al. (2019)[23] provided a critical analysis of the summarization field, emphasizing limitations related to data quality. Gupta et al. (2021)[24] explored transformer-based summarization of news data, while another study by Gupta et al. (2022)[25] implemented ranking-based models to improve accuracy. Yadav et al. (2022)[26] demonstrated the use of deep learning architectures, and Pal et al. (2022)[27] compared the performance of models like BERT and T5. Usha Manjari et al.[28] extended extractive summarization research to include web-based content, demonstrating the flexibility of these models across domains.

Kore et al.'s[29] approach integrated machine learning algorithms such as SVM and decision trees with NLP techniques like TF-IDF and word embeddings to enhance both sentence relevance and summary cohesion. Similarly, Ghosh et al.[30] emphasized the impact of pre-processing techniques—such as stopword elimination, lemmatization, and named entity recognition—on improving both extractive and abstractive summarization performance in Indian legal texts.

Other new methods include the work of Megala et al. [31], who created a mixed system that uses fuzzy logic along with Conditional Random Fields (CRF) to rank and pick sentences based on language and context clues. Dixit[32] built on this fuzzy logic idea, showing how it can effectively show the importance of sentences by looking at how similar their meanings are. Onan and Alhumyani[33] made FuzzyTP-BERT, which combines fuzzy topic modeling with transformer-based language models to make better summaries by matching deep word meanings with topic relevance.

Particle Swarm Optimization (PSO) has also found wide application in text processing. Chang (2021)[34] proposed a refined PSO algorithm for constrained nonlinear programming, laying groundwork for advanced optimization applications. In the same year, Fregoso et al.[35] leveraged PSO to optimize CNN configurations for sign language recognition, underscoring its value in deep learning optimization. Alhaj et al. (2022)[36] applied an improved PSO technique for Arabic text classification, adapting it to language-specific features. Ali et al.[37] used PSO to enhance Arabic named entity recognition from social media, demonstrating its resilience in processing noisy and informal datasets. Ezaldeen et al.[38] incorporated normalized PSO into a context-aware learner model for adaptive e-learning, showcasing its use in personalized education systems. Aote et al. (2023)[39] combined binary PSO with a genetic algorithm for multi-document summarization[40] in Hindi, highlighting the potential of evolutionary strategies in enhancing summarization outcomes across languages and domains.

## IV Dataset

To thoroughly assess the effectiveness of our proposed summarization approach, we employed three diverse datasets, each catering to different aspects of legal and multi-document summarization tasks.

### *i Indian Legal Corpus*

- This dataset[41] has 3,122 rulings from important Indian courts.
- Each document contain 8,000 words on average.
- Legal experts have written judgment with summaries in both English and Hindi languages.
- One of the first large resources made for summarizing legal texts in different languages in India.

### *ii MILDSum Dataset*

- MILDSum[42] is a special dataset made to summarize the legal text. It contains 3,000+ legal articles.
- Each document contains about 4,700 tokens, while the each summary contains 724 tokens in English and 695 in Hindi on average.
- The dataset is useful for testing summarization tools in the legal area.
- It is especially helpful for testing models on multilingual summarization tasks and dealing with complex legal documents.

### *iii DUC2004 Dataset*

- The DUC 2004 dataset[43] is a standard dataset used for document summarization which was released during the Document Understanding Conference, .
- It contains 500 news articles, each articles is associated with four summaries written by humans.
- Each article length is between 800 to 1,000 words on average and these articles are grouped into 50 clusters, with about 10 articles in each cluster.
- The articles come from major news sources such as AP Newswire (1998–2000), New York Times Newswire (1998–2000), and the English version of Xinhua News Agency (1996–2000).

## V Methodology

This research aims at an improved method for extractive summarization that focuses on integrating optimization technique with the statistical methods, to improve the final set of sentences, in order to generate high quality summaries. The whole process involves calculating the importance of each sentence, using a swarm-based algorithm to optimize the sentence selection, and applying Frequency Domain analysis for the final touch.

### *i Sentence Scoring Using TF-IDF*

As first step, TF-IDF is used to measure how important each sentence is based on how often words appear in the document. Let's say an article has sentences $S_1, S_2, \ldots, S_n$, and each sentence $S_i$ contains words $W_{i1}, W_{i2}, \ldots, W_{im}$. The TF-IDF score for each word is calculated as:

$$\text{TF-IDF}(W_{ij}) = \text{TF}(W_{ij}) \times \text{IDF}(W_{ij})$$

Where: - $\text{TF}(W_{ij})$: The frequency of the word $W_{ij}$ in the sentence $S_i$.

$\text{IDF}(W_{ij}) = \log\left(\frac{N}{df(W_{ij})}\right)$, with $N$ as the total number of sentences and $df(W_{ij})$ as the number of sentences containing $W_{ij}$.

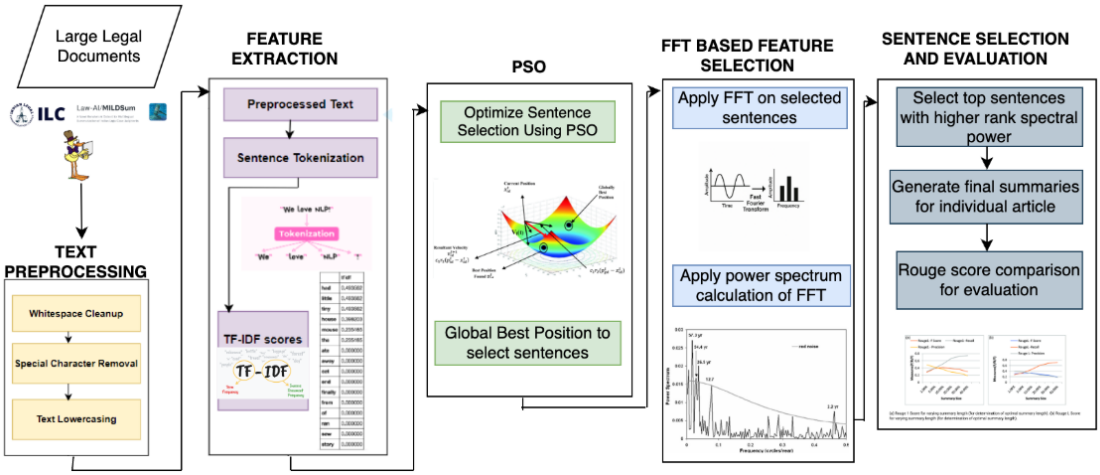


Figure 3: Overall Architecture of Proposed Methodology.

The sentence score is computed as the sum of TF-IDF scores of its constituent words:

$$ss_i = \sum_{j=1}^{m} \text{TF-IDF}(W_{ij})$$

These sentence scores act as the basis for optimization in the subsequent phase.

## *ii Sentence Selection Using Particle Swarm Optimization (PSO)*

PSO is employed to identify the most informative subset of sentences while maintaining a balance between content richness and conciseness.

### ii.1 Initialization

- **Particle Encoding:** Each particle represents a potential summary and is encoded as a binary vector, where 1 indicates inclusion and 0 exclusion of a sentence.
- **Position and Velocity:** Initial particle positions are randomly generated, and velocities are initialized to zero.

### ii.2 Fitness Evaluation

A fitness function evaluates the quality of each candidate summary by aggregating the sentence scores:

$$\text{Fitness} = \sum(\text{Particle Position} \times ss)$$

This rewards summaries composed of higher-scoring sentences.

### ii.3 Iterative Optimization Process

For each iteration:

1. **Update Best Positions:** Each particle updates its personal best and compares with the global best.

2. **Velocity Update:**

$$v = w \cdot v + c_1 \cdot r_1 \cdot (\text{pBest} - \text{Current Position}) + c_2 \cdot r_2 \cdot (\text{gBest} - \text{Current Position})$$

where $w$ is inertia weight, $c_1, c_2$ are learning coefficients, and $r_1, r_2$ are random numbers in [0,1].

3. **Position Update:**

$$\text{New Position} = \text{Current Position} + v$$

4. **Final Selection:** After the last iteration, the global best particle yields the selected sentences.

### *iii Improving with Fast Fourier Transform (FFT)*

FFT is used to look at the selected sentences in a different way, helping to find which ones are most important based on their structure and meaning.

#### iii.1 Changing to Frequency Domain

For each sentence score $ss_j$, we apply FFT:

$$\text{fft}_j = |\text{FFT}(ss_j)|$$

This helps show how important each part of the sentence is across different frequencies.

#### iii.2 Calculating Power Spectrum

We then calculate the power spectrum to see how important each sentence really is:

$$ps_j = \frac{\text{fft}_j^2}{t}$$

where $t$ is the total number of sentences. The higher the $ps_j$, the more important the sentence is.

#### iii.3 Final Ranking of Sentences

The sentences are sorted from most important to least, and the top ones are picked to create the summary.

### *iv Summary Construction*

The final summary is formed by combining the highest-ranked sentences—selected through PSO and refined via FFT. This hybrid strategy merges statistical scoring, heuristic optimization, and frequency-based analysis to produce concise, informative, and structurally coherent summaries suitable for long and complex documents.

## VI Results

This section presents a detailed performance evaluation of the proposed summarization model across three benchmark datasets—DUC 2004, ILC, and MILDSUM. The evaluation is based on key metrics such as ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore (where applicable), to assess both the content overlap and semantic quality of the generated summaries. Additionally, we report average processing times to highlight the efficiency of the proposed approach.

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**Algorithm 1** Summarization Algorithm using TF-IDF, PSO, and FFT

**Input:** Articles $A_1, A_2, ..., A_k \in A$.
**Output:** Summaries $S_1, S_2, ..., S_k$.

**Initialization:** - Total number of articles: $k$ - Words per article: $W$ - Total words: $l$ - Sentence scores: $ss$ - FFT scores: $fft$ - Power spectrum: $ps$ - Selected indices: $I$
**PSO Parameters:** - Particles: $P$ - Iterations: $I$ - Inertia weight: $w$ - Cognitive/social constants: $c_1, c_2$

1. **Compute Sentence Scores Using TF-IDF:**
   (a) For each article $A_i$, compute TF-IDF scores for each sentence $S_j$.
   (b) Store sentence scores:
   $$ss_j = [\text{TF-IDF}(W_1), \ldots, \text{TF-IDF}(W_m)]$$
2. **Optimize Sentence Selection Using PSO:**
   (a) Initialize $P$ particles with random binary positions and velocity $v = 0$.
   (b) **For** each iteration from 1 to $I$:
      i. Compute fitness:
      $$f = \sum(\text{Particle Position} \times ss)$$
      ii. Update particle best and global best positions.
      iii. Update velocity:
      $$v = w \cdot v + c_1 \cdot (\text{Particle Best}) + c_2 \cdot (\text{Global Best})$$
      iv. Update position:
      $$\text{Position} = \text{Position} + v$$
   (c) Select top $n$ positions from the global best.
3. **Apply FFT to Selected Sentences:**
   (a) Compute FFT for each selected sentence:
   $$fft_j = |\text{FFT}(ss_j)|$$
   (b) Compute power spectrum:
   $$ps_j = fft_j^2$$
4. **Generate Summary:**
   (a) Sort $ps$ in descending order.
   (b) Select top $n$ sentences.
   (c) Append selected sentences to $S_i$.

## *i DUC 2004 Dataset Evaluation*

The DUC 2004 dataset is widely regarded as a standard benchmark for automatic text summarization. We compared our model with established methods including LexRank, TextRank, MMR, and several Pointer-Generator (PG) based techniques.

### i.1 ROUGE-1

Our hybrid summarization method achieved a ROUGE-1 score of 57.1, outperforming baseline models such as LexRank (35.56) and TextRank (33.16)[45]. This indicates a stronger ability to capture unigram-level information, making the summaries more representative of the source documents.

### i.2 ROUGE-2

For ROUGE-2, the model secured a score of 12.1, significantly exceeding results from PG-MMR (9.36) and Hi-MAP (8.90). This suggests improved preservation of word pairings and phrase-level meaning in the generated summaries.

### i.3 ROUGE-L

Our method recorded a ROUGE-L score of 27.1, which surpassed LexRank (24.4) and TextRank (25.8). This reflects better handling of longer textual dependencies and more coherent summary structures.

Table 1: Comparison on DUC2004 (Tomer and Kumar, Hark and Karci)[38][39]

| **Method** | **Rouge 1** | | | **Rouge 2** | | | **Rouge L** | | |
|---|---|---|---|---|---|---|---|---|---|
| | P | R | F | P | R | F | P | R | F |
| LSA | .285 | .313 | .297 | .038 | .041 | .039 | .238 | .248 | .260 |
| TextRank | .334 | .398 | .362 | .068 | .079 | .073 | .258 | .306 | .279 |
| LexRank | .304 | .322 | .312 | .051 | .054 | .052 | .244 | .257 | .250 |
| RBM | .238 | .273 | .254 | .071 | .089 | .082 | .210 | .241 | .423 |
| FLSTM | .518 | .480 | .456 | .077 | .154 | .242 | .350 | .345 | .464 |
| SumBasic | .318 | .338 | .328 | .052 | .056 | .054 | .236 | .244 | .236 |
| Karci | .345 | .391 | .366 | .073 | .082 | .077 | .265 | .301 | .281 |
| **Proposed** | **.571** | .175 | .265 | **.121** | .037 | .056 | **.271** | .083 | .126 |

Table 2: Comparison on DUC2004 (other methods)

| **Method** | **R1** | **R2** |
|---|---|---|
| LexRank | 35.56 | 7.87 |
| TextRank | 33.16 | 6.13 |
| MMR | 30.14 | 4.55 |
| PG-Original | 31.43 | 6.03 |
| PG-MMR | 36.42 | 9.36 |
| PG-BRNN | 29.47 | 6.77 |
| CopyTransformer | 28.54 | 6.38 |
| Hi-MAP | 35.78 | 8.90 |
| **Proposed-Method** | **57.1** | **12.1** |

## *ii ILC Dataset Evaluation*

The Indian Legal Corpus (ILC) was used to further test the model's robustness in legal domain summarization. We focused on F1 and recall scores across ROUGE-1, ROUGE-2, and ROUGE-L.

### ii.1 ROUGE-1

Our model achieved a ROUGE-1 F1 score of 38.47, higher than TextRank (34.63) and LexRank (33.21), highlighting its stronger performance in extracting critical content from long legal documents.

### ii.2 ROUGE-2

The ROUGE-2 F1 score reached 22.12, ahead of both TextRank (18.16) and LexRank (16.94), showing its effectiveness in maintaining contextual relationships between word pairs.

### ii.3 ROUGE-L

For ROUGE-L, our method scored 36.39, outperforming other models including TextRank (31.11) and LexRank (30.12), demonstrating better preservation of sentence structure and document flow.

Moreover, the recall scores were particularly strong: ROUGE-1 recall: 42.88 ROUGE-2 recall: 25.27 ROUGE-L recall: 40.49 These results confirm that the model is not only accurate but also comprehensive in retaining key information.

Table 3: Comparison on ILC Dataset[37]

| **Method** | **Rouge 1** | | **Rouge 2** | | **Rouge L** | |
|---|---|---|---|---|---|---|
| | F1 | Recall | F1 | Recall | F1 | Recall |
| Sumbasic | 15.69 | 9.5 | 6.02 | 3.5 | 14.48 | 9 |
| LSA | 21.20 | 14.08 | 7.37 | 4.6 | 19.76 | 13.11 |
| KLSum | 21.40 | 13.39 | 10.19 | 6.15 | 19.66 | 12.28 |
| LexRank | 33.21 | 25.60 | 16.94 | 12.65 | 30.12 | 23.18 |
| TextRank | 34.63 | 28.70 | 18.16 | 14.62 | 31.11 | 25.79 |
| LED | 4.31 | 2.28 | 1.08 | 0.5 | 4.11 | 2.17 |
| **Proposed Method** | **38.47** | **42.88** | **22.12** | **25.27** | **36.39** | **40.49** |

## *iii MILDSUM Dataset Evaluation*

The MILDSUM dataset, designed for extractive summarization in legal settings, was evaluated using ROUGE-L and BERTScore.

### iii.1 ROUGE-L

Our model produced a ROUGE-L score of 33.4, comparable to leading methods such as SummaRuNNer (30.34) and Legal-Pegasus (28.36). This indicates the model's effectiveness in generating summaries with coherent sentence-level structure.

### iii.2 BertScore

In terms of semantic similarity, the model achieved a BERTScore of 82.28, slightly lower than fine-tuned transformer-based models like LED (84.20), but still within a competitive range. This suggests the summaries align well with ground truth in terms of meaning and relevance.

### *iv Processing Time and Efficiency*

Efficiency is vital for practical summarization systems. Our model demonstrated favorable average processing times:

DUC 2004: 26.84 seconds per article (approx. 1000 tokens)

ILC (LegalText): 17.67 seconds per article (approx. 8000 tokens)

Despite its strong performance, the model remains computationally efficient, making it suitable for real-time applications. Unlike transformer-based models like BERT, which require significant computational power and model training, our approach is training-free, eliminating the need for large datasets, GPUs, or prolonged optimization cycles.

Its lightweight and deterministic design ensures fast execution and scalability, making it especially ideal for low-resource settings or time-sensitive environments.

| **Document** | **Number of Words per Document** | **Processing Time per Document (seconds)** |
|---|---|---|
| DUC 2004 | ~1,000 | 1.70 |
| LegalText (ILC) | ~8,000 | 17.67 |
| MILDSUM | ~4,000 | 8.16 |

Table 4: Processing time and document length for different datasets

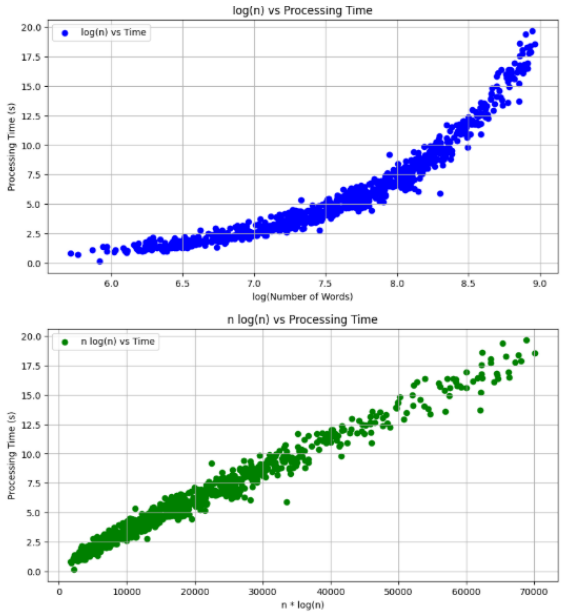



Figure 4: Length of Article vs processing time

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## VII Conclusion

This research proposed a new way to generate efficient summaries by combining Particle Swarm Optimization (PSO) with Fast Fourier Transform (FFT). The aim is to make the process of choosing and improving summary sentences more effective. PSO helps find the best sentences to include, while FFT looks at how important each sentence is by analyzing them in a different way. Using both methods together made the summaries more informative.

The approach was tested on three different sets of data: DUC 2004, ILC, and MILDSUM. It performed better than the summarization tools traditionally used. The model showed great results in ROUGE-1, ROUGE-2, and

Table 5: Comparison on MILDSUM Dataset[40]

| **Method** | **Rouge 2** | **Rouge L** | **BERT Score** |
|---|---|---|---|
| LexRank | 30.49 | 29.24 | 83.97 |
| LSA-based | 29.46 | 28.54 | 83.71 |
| Luhn | 28.76 | 28.12 | 83.58 |
| SummaRuNNer | 32.27 | 30.34 | 84.13 |
| BertSumExt | 28.71 | 29.11 | 83.77 |
| Legal-Pegasus-finetuned | 31.84 | 28.36 | 84.14 |
| Long T5-finetuned | 23.25 | 23.47 | 83.90 |
| LED-finetuned | 25.51 | 27.81 | 84.20 |
| **Proposed Method** | 18.69 | **33.4** | 82.28 |

ROUGE-L scores. Summaries thus generated kept more information, handled longer phrases better, and had a smoother flow.

Model was also efficient in speed. Since no training is required, it made a good balance between the speed and the quality with which the summaries were generated making it highly useful for real-world use cases, especially when dealing with long or complicated legal or technical files.

However, there are few areas that the research looks forward to improving. To make the process even faster, mixing it with deep learning methods and adapting it to more languages could make the model even better. These changes could help the model work well in bigger and more varied summarization tasks.

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